

Intro to the idea of a confidence interval

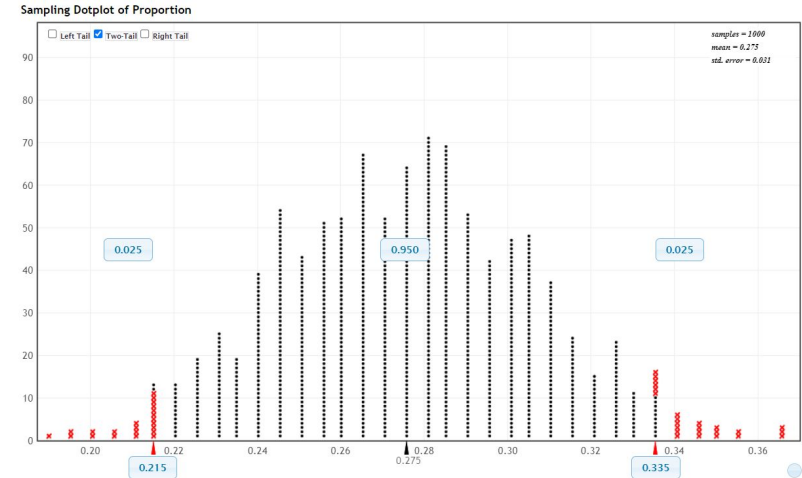
MATH 213

Group 1

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?
2. What does each dot represent?

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

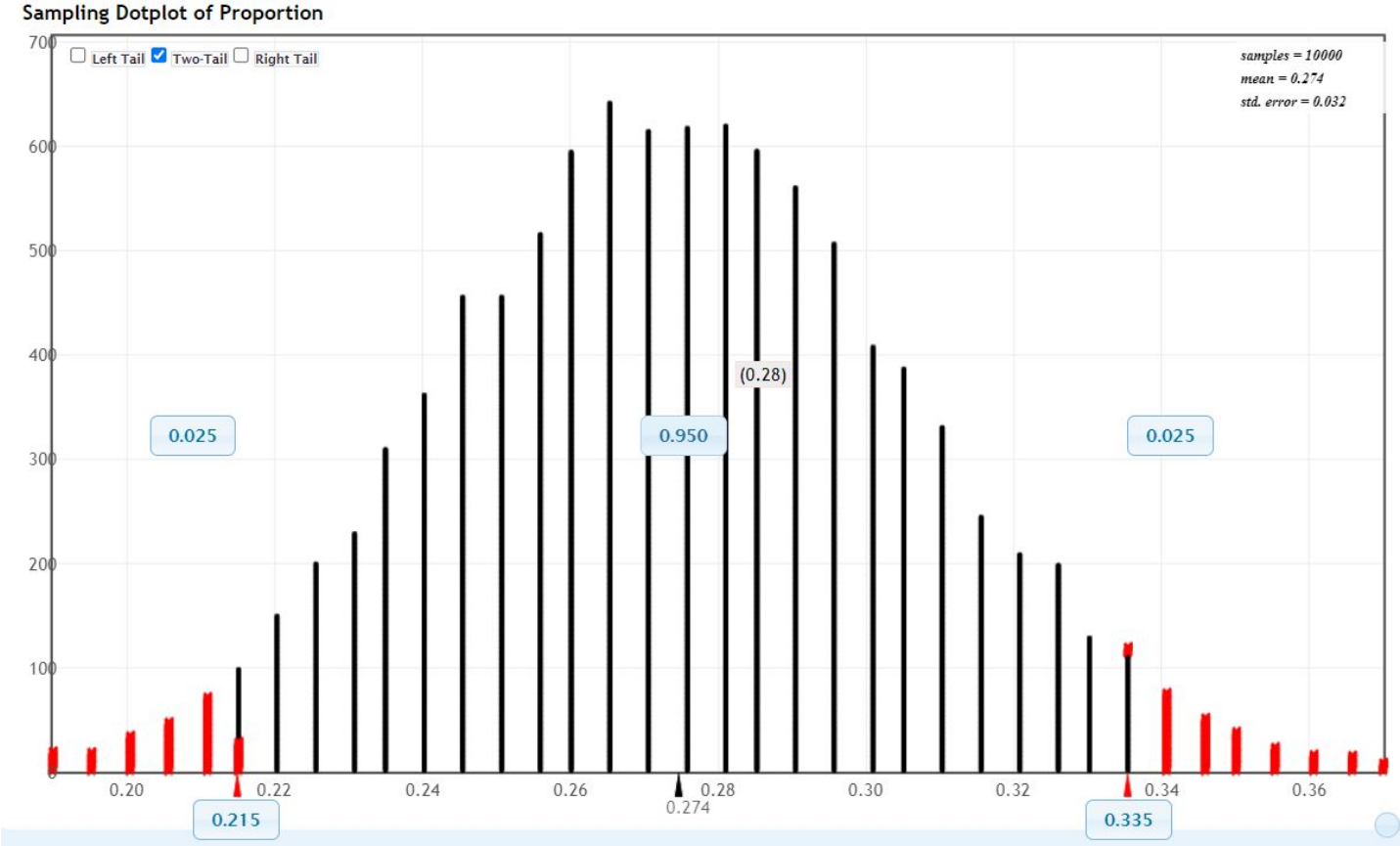
Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
4. What percentage of dots lies in the red area on the right tail?
5. What percentage of dots lies in the black area in the middle?
7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

Fill in the blanks:

8. _____% of all sample proportions lie within a distance of _____ from the population proportion.
9. _____% of all sample proportions lie further than _____ away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

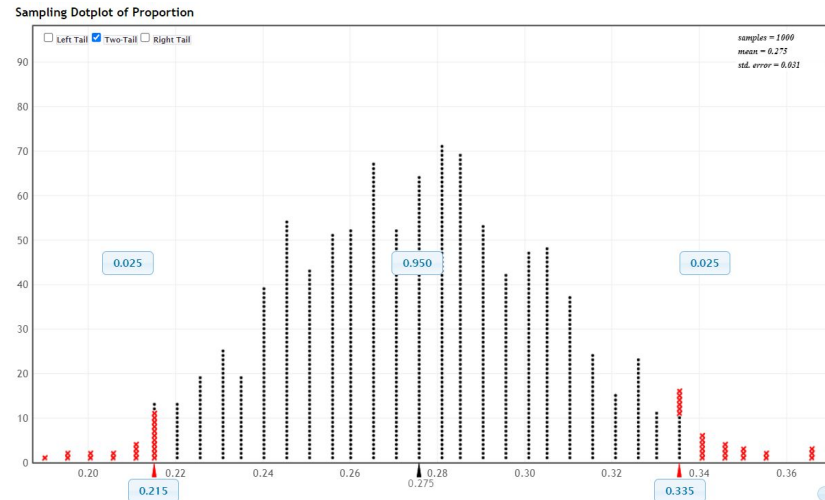


Group 2

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?
1000
2. What does each dot represent?
Each dot represents the proportion of a 200-sized sample that has a college degree.

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
2.5%
4. What percentage of dots lies in the red area on the right tail?
2.5%
5. What percentage of dots lies in the black area in the middle?
95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

.06

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

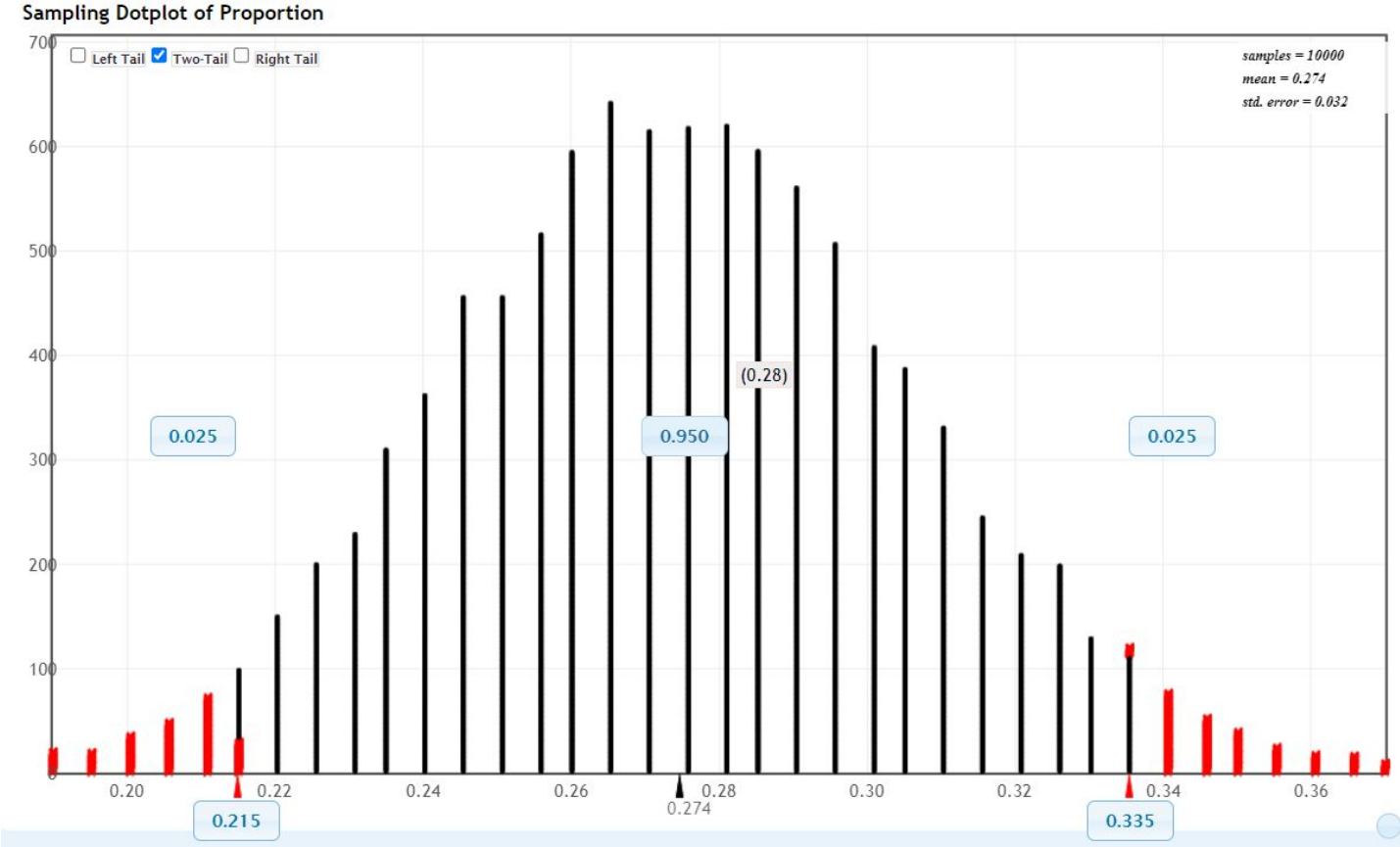
.06

Fill in the blanks:

8. 95?% of all sample proportions lie within a distance of .06 from the population proportion.

9. 5% of all sample proportions lie further than .06 away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

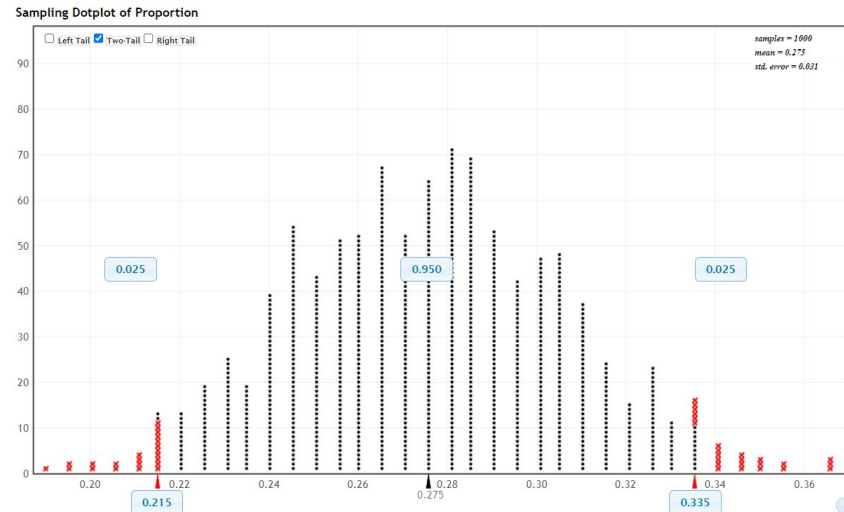


Group 3

From the most recent census:

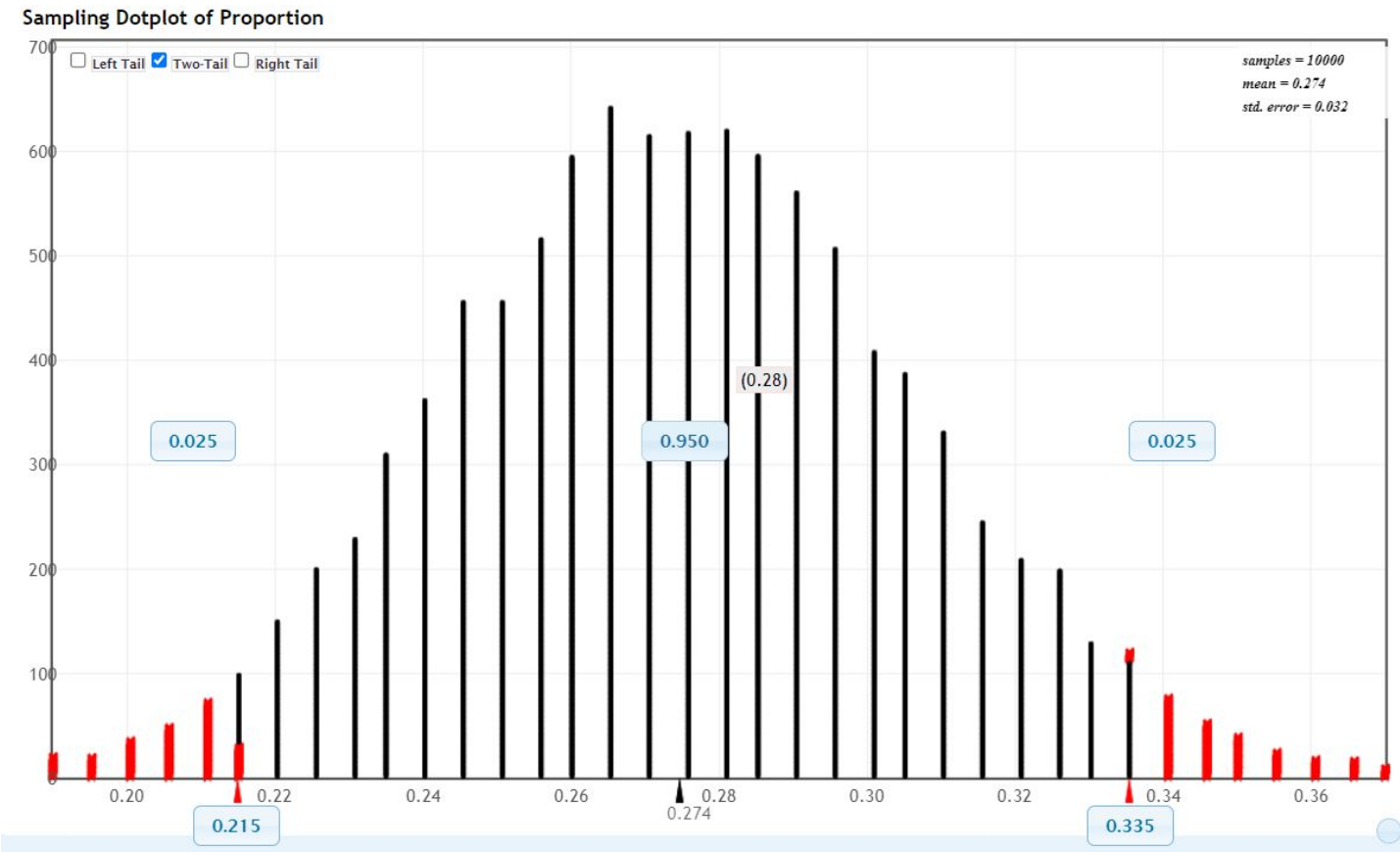
27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.



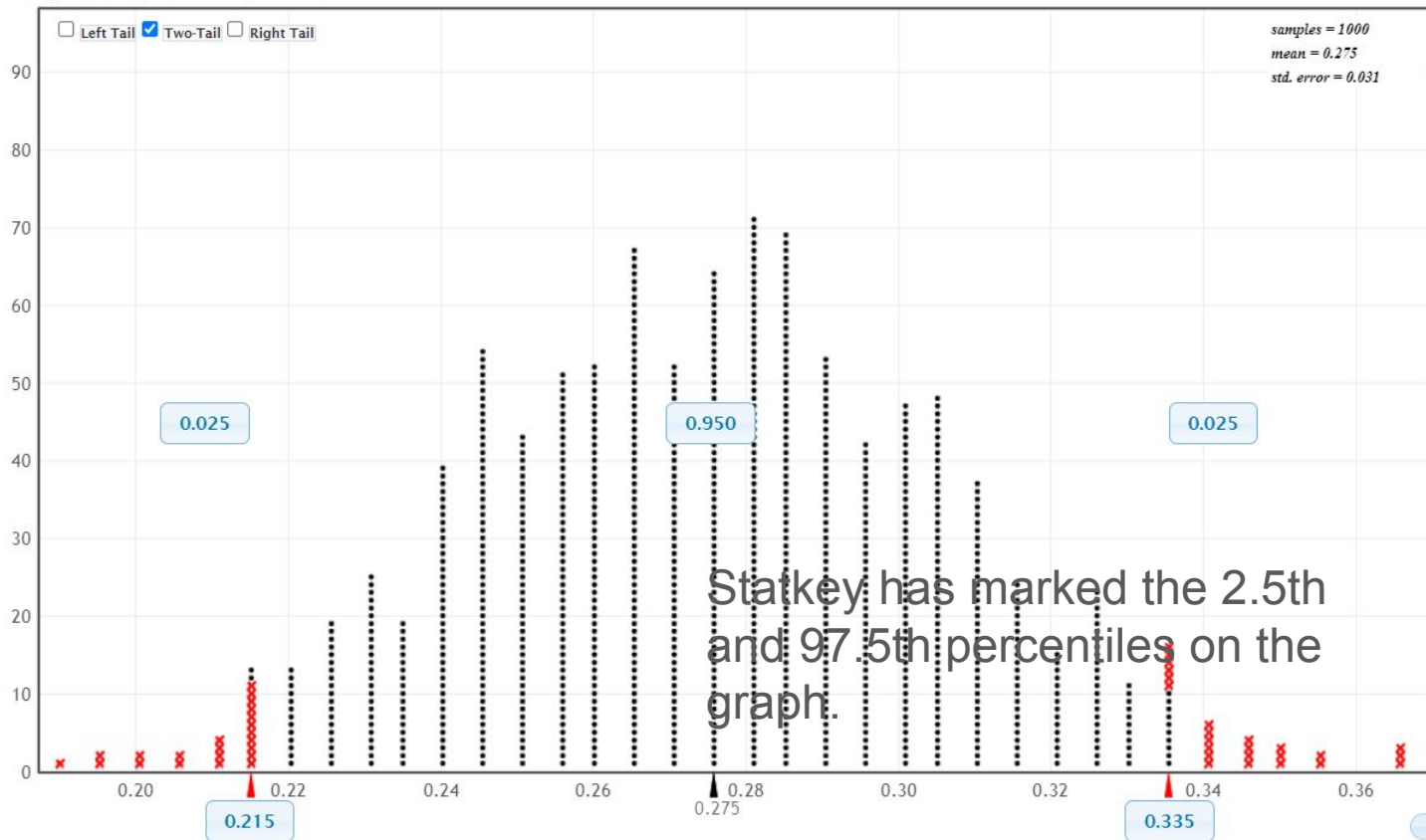
Group 4

From the most recent ce

27.5% of all US resider college degree.

- This 27.5% is a pa proportion.
- I used StatKey to (sample size $n = 20$)
 - Select a sam calculate the at least a col
 - Do this over
 - This is the s
 - Features: Ap
 - Centered at spread is als a formula for

Sampling Dotplot of Proportion



1. How many dots on this dotplot?

1000

2. What does each dot represent?

A sample proportion of US adults that have a college degree

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?

2.5%

4. What percentage of dots lies in the red area on the right tail?

2.5%

5. What percentage of dots lies in the black area in the middle?

95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

).120

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

0.060

Fill in the blanks:

8. _____% of all sample proportions lie within a distance of _____ from the population proportion.

9. _____% of all sample proportions lie further than _____ away from the population proportion.

1. How many dots on this dotplot?
 - 1000
2. What does each dot represent?
 - The proportion of the sample with a college degree for each simulation of 200 samples

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
 - 2.5%
4. What percentage of dots lies in the red area on the right tail?
 - 2.5%
5. What percentage of dots lies in the black area in the middle?
 - 95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

.060

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

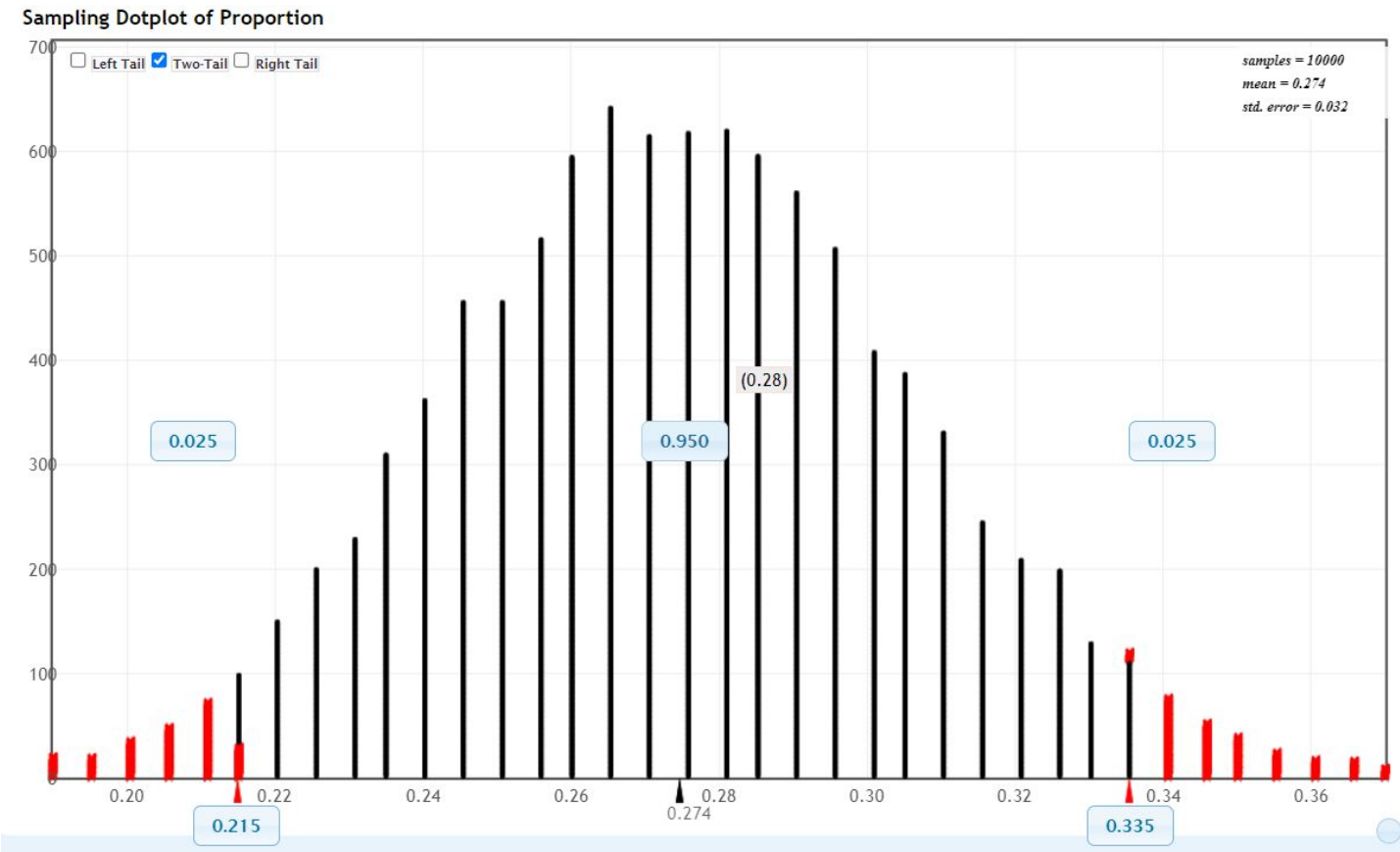
.060

Fill in the blanks:

8. 95% of all sample proportions lie within a distance of .060 from the population proportion.

9. 5% of all sample proportions lie further than .060 away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

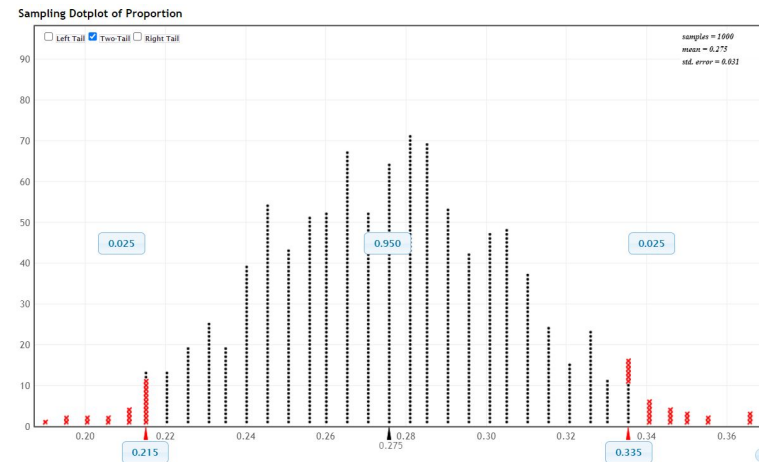


Group 5

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?
1000
2. What does each dot represent?

A proportion of the sample with a sample size of 200

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
0.025 → 2.5%
4. What percentage of dots lies in the red area on the right tail?
0.025 → 2.5%
5. What percentage of dots lies in the black area in the middle?
0.950 → 95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

0.06

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

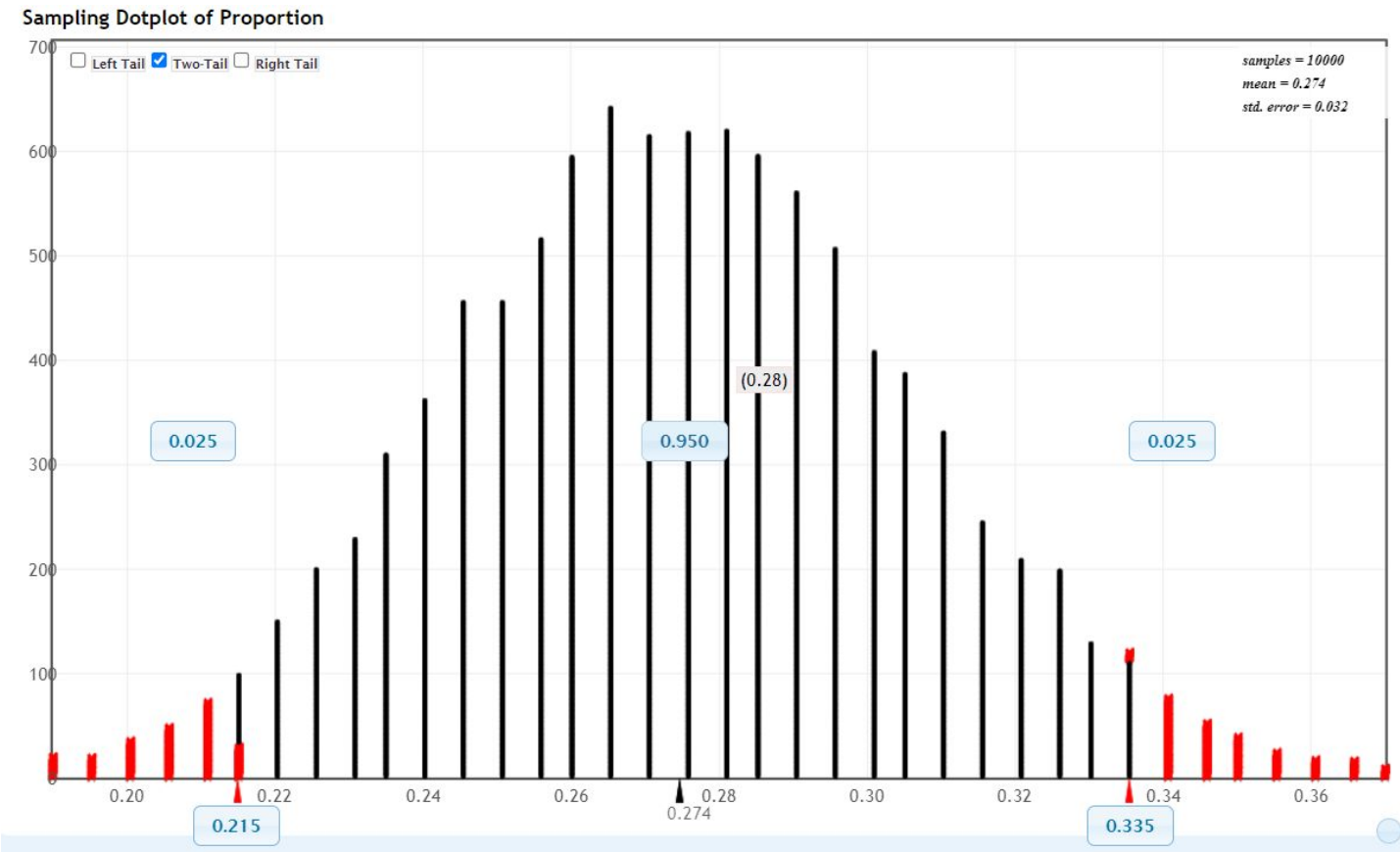
0.06

Fill in the blanks:

8. **95%** of all sample proportions lie within a distance of **0.06** from the population proportion.

9. **5%** of all sample proportions lie further than **0.06** away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.



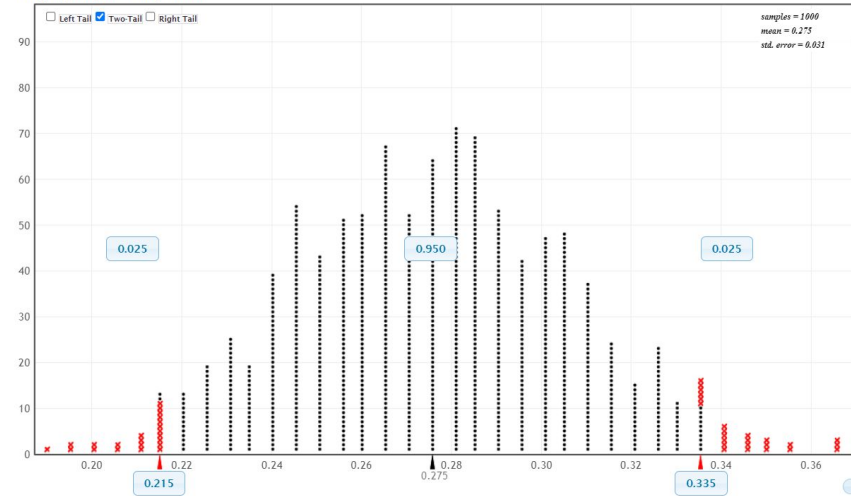
Group 6

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.

Sampling Dotplot of Proportion



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot? 1000

2. What does each dot represent?

The proportion of a sample of 200 who have a college degree

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?

2.5

4. What percentage of dots lies in the red area on the right tail?

2.5

5. What percentage of dots lies in the black area in the middle?

95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

$$.335 - .275 = .06$$

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

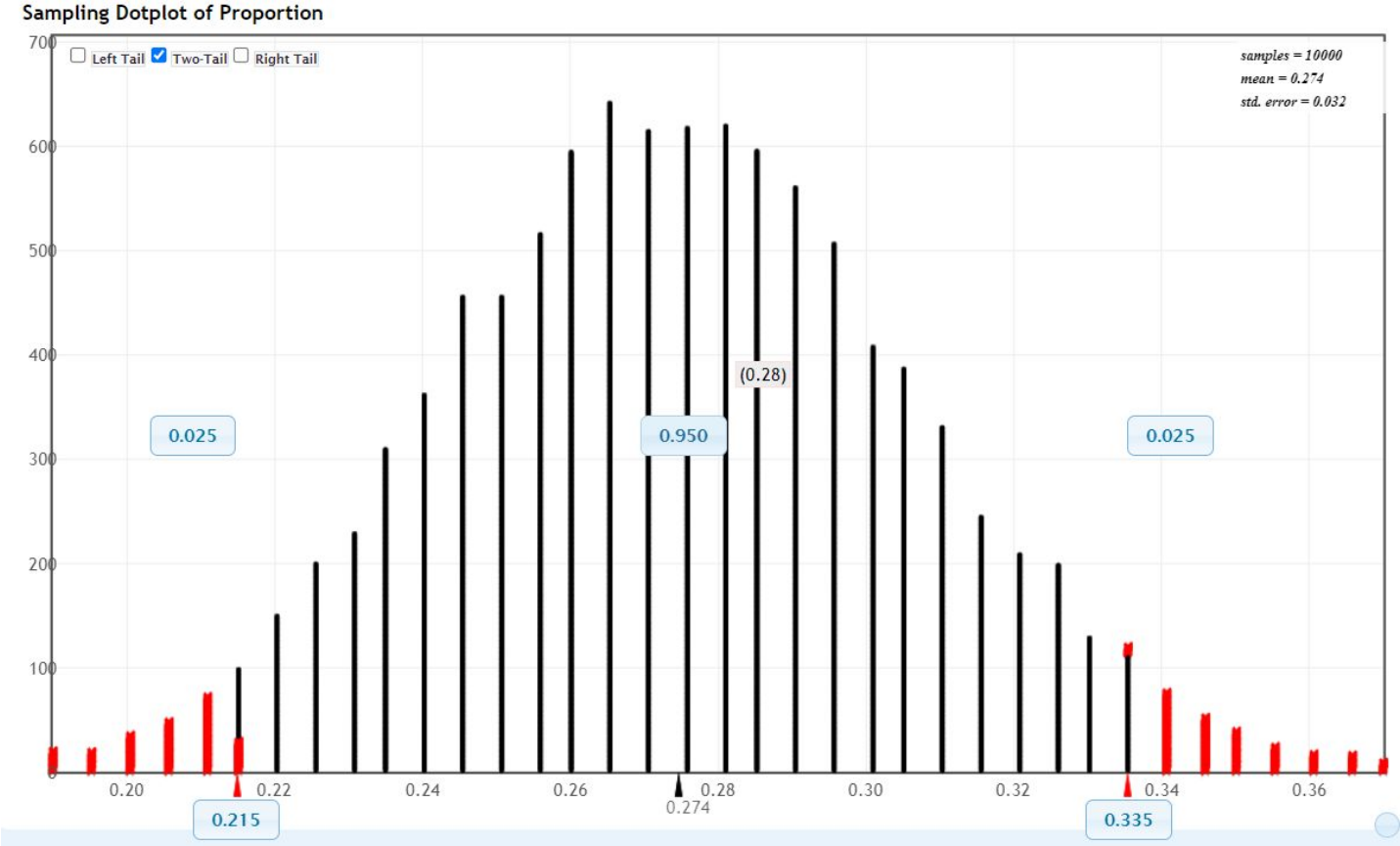
$$.275 - .215 = .06$$

Fill in the blanks:

8. ____2.5__% of all sample proportions lie within a distance of ____0.06__ from the population proportion.

9. ____2.5____% of all sample proportions lie further than ____0.06____ away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

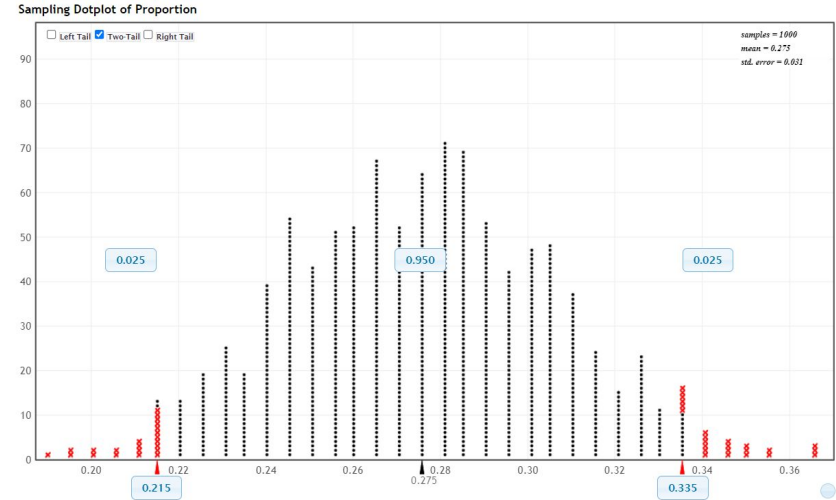


Group 7

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?

1000

2. What does each dot represent?

The proportion of people who have a college degree in a sample of 200

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?

2.5%

4. What percentage of dots lies in the red area on the right tail?

2.5%

5. What percentage of dots lies in the black area in the middle?

95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

0.475

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

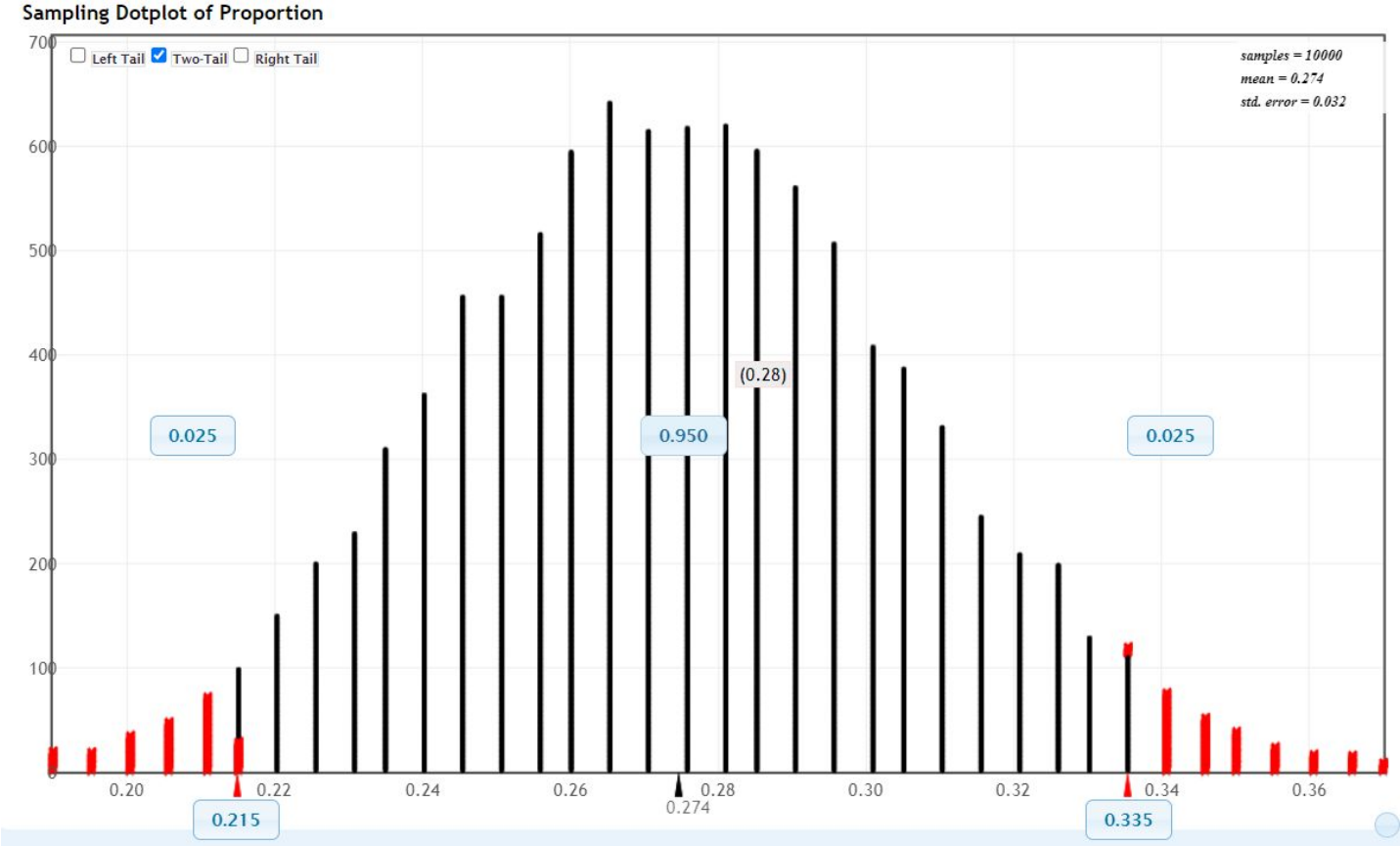
0.475

Fill in the blanks:

8. 95% of all sample proportions lie within a distance of 0.475 from the population proportion.

9. 5% of all sample proportions lie further than 0.475 away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

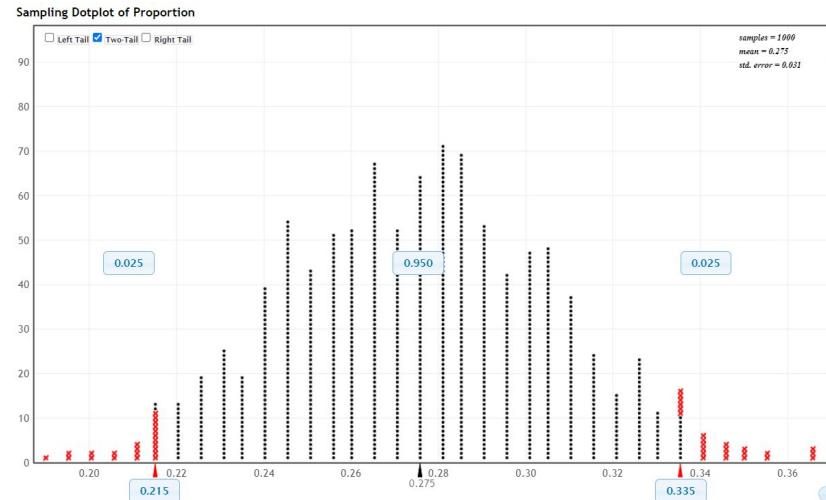


Group 8

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?

200

2. What does each dot represent?

It represents the number of people who have at least a college degree

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?

2.5

4. What percentage of dots lies in the red area on the right tail?

2.5

5. What percentage of dots lies in the black area in the middle?

95.0

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

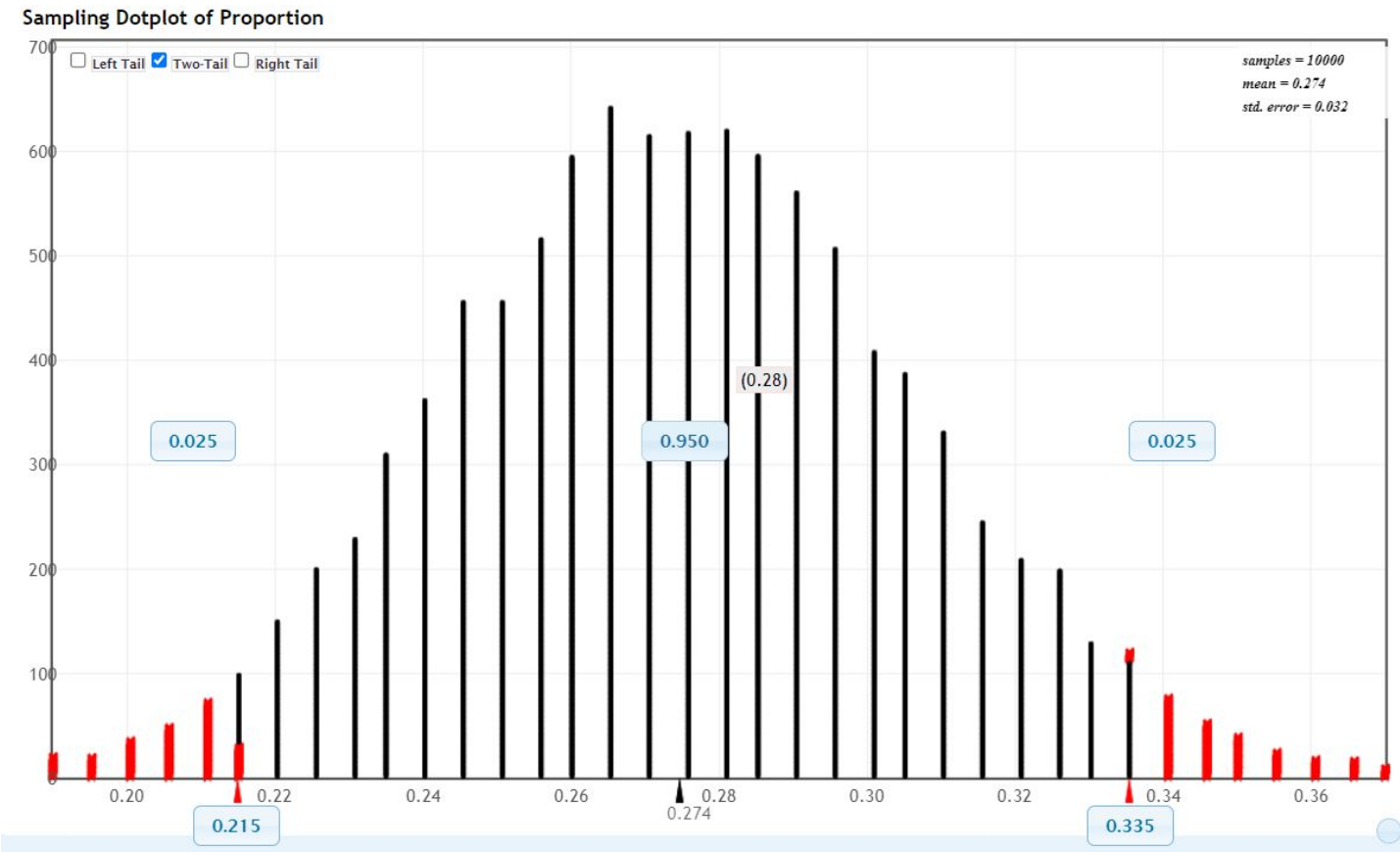
7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

Fill in the blanks:

8. _____% of all sample proportions lie within a distance of _____ from the population proportion.

9. _____% of all sample proportions lie further than _____ away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

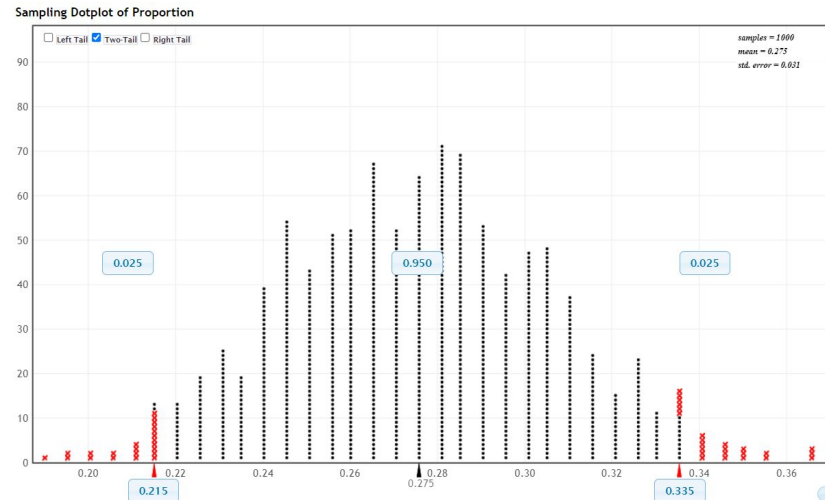


Group 9

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?
1000

2. What does each dot represent?

The percentage of a randomly sampled group of 200 people that are US residents aged 25 and older from the total population that have a college degree

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
2.5%

4. What percentage of dots lies in the red area on the right tail?
2.5%

5. What percentage of dots lies in the black area in the middle?
95%

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

0.12

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

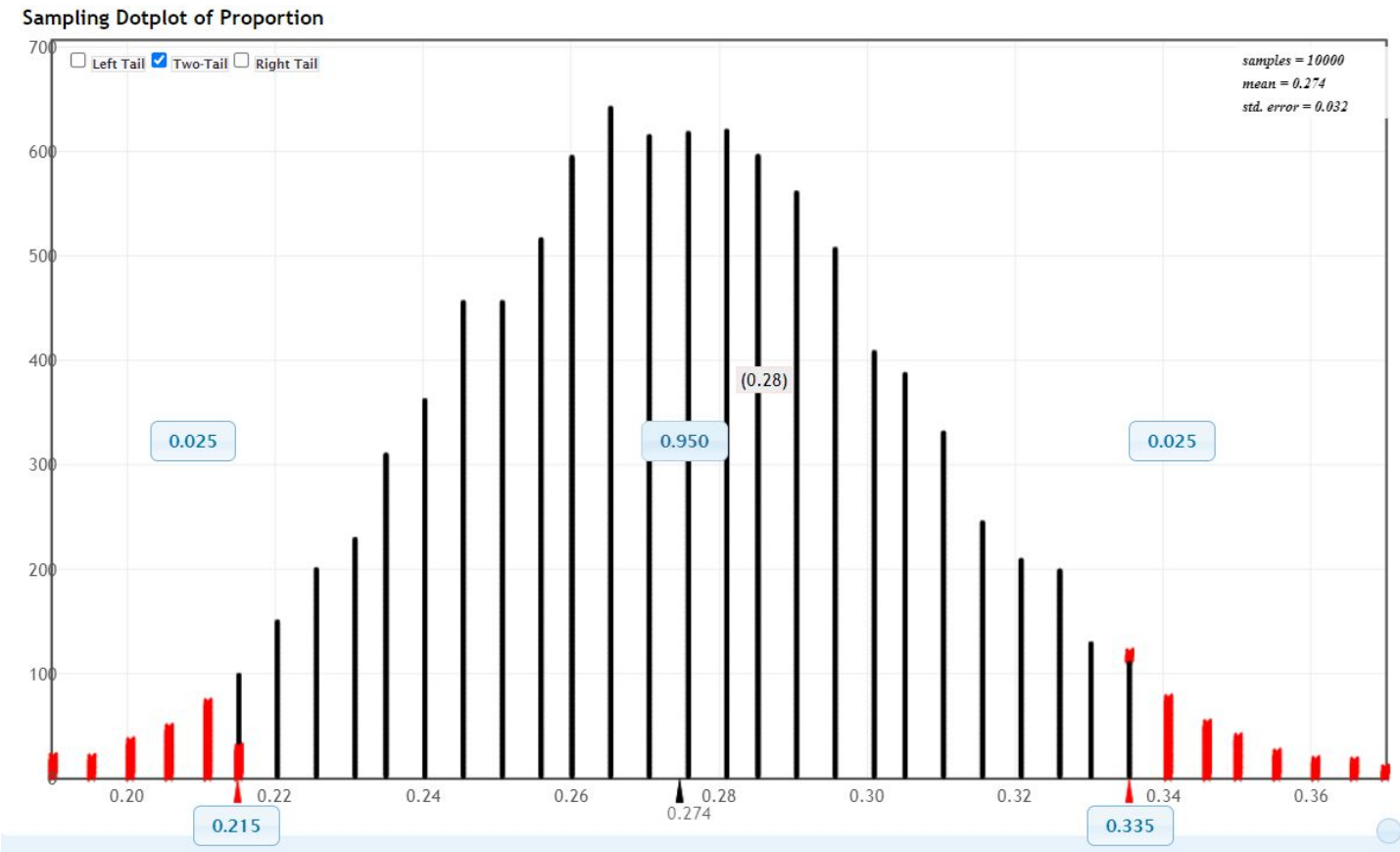
0.06

Fill in the blanks:

8. **95%** of all sample proportions lie within a distance of **0.06** from the population proportion.

9. **5%** of all sample proportions lie further than **0.06** away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

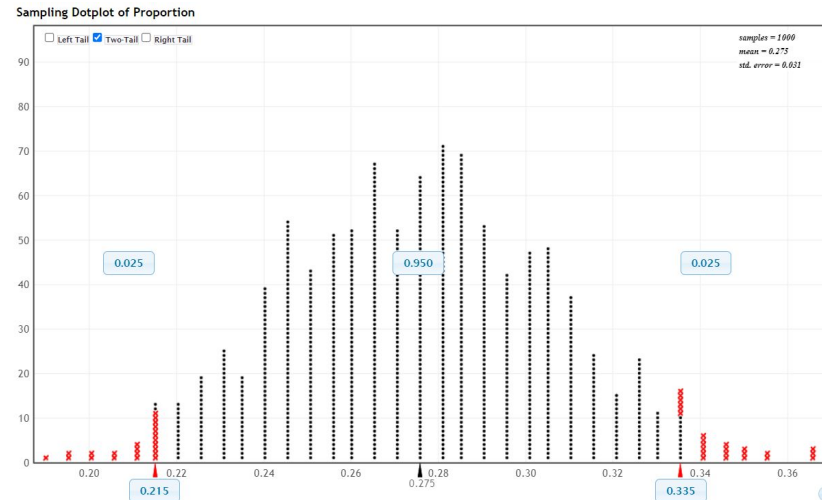


Group 10

From the most recent census:

27.5% of all US residents aged 25 and older have a college degree.

- This 27.5% is a parameter; specifically a population proportion.
- I used StatKey to construct a sampling distribution with sample size $n = 200$.
 - Select a sample of size 200 from the population, calculate the proportion of that sample who has at least a college degree, plot it on the dotplot.
 - Do this over and over again 1000 times.
 - This is the sampling distribution.
 - Features: Approximately symmetric, bell-shaped. Centered at 0.275, the population proportion. The spread is also predictable though we don't have a formula for it right now.



Statkey has marked the 2.5th and 97.5th percentiles on the graph.

1. How many dots on this dotplot?
2. What does each dot represent?

Statkey marked the 2.5th and 97.5th percentiles.

3. What percentage of the dots lies in the red area on the left tail?
4. What percentage of dots lies in the red area on the right tail?
5. What percentage of dots lies in the black area in the middle?

6. What is the distance from the center of the sampling distribution (the population proportion of 0.275) to the value at the 97.5th percentile? Note: *distance is always positive*.

7. What is the distance between the center of the distribution and the value at the 2.5th percentile?

Fill in the blanks:

8. _____% of all sample proportions lie within a distance of _____ from the population proportion.

9. _____% of all sample proportions lie further than _____ away from the population proportion.

Notice: if I select more and more samples, the percentage of samples whose sample proportions lie within about 0.06 of the population parameter does not change. Here is a sampling distribution that has 10,000 sample proportions displayed.

